

PROJECT DOCUMENTATION

BRAINSTORMING AND IDEATION

EduGenie – Google Gemini Powered Learning Assistant

1. Project Idea

EduGenie is an AI-powered educational learning assistant designed to help students learn concepts, ask academic questions, generate quizzes, summarize learning material, and receive personalized learning guidance. The project uses Google Gemini as the generative AI engine, FastAPI as the backend framework, Python for application logic, and HTML/CSS with Jinja2 templates for the web interface.

2. Need Identification

Students may need immediate explanations, practice questions, summaries, and structured guidance. Traditional resources such as textbooks, search engines, videos, and separate study tools require learners to organize information manually.

3. Brainstormed Features

- Ask questions and receive concise AI answers.
- Explain difficult concepts in simple language.
- Generate quizzes from topics or text.
- Summarize educational passages.
- Create personalized learning paths with topics and timelines.

4. Proposed Solution

A single AI-powered web interface combines common learning activities. The student enters a question or topic, the FastAPI backend communicates with Google Gemini, and the generated educational response is displayed through the EduGenie webpage.

5. Technology Ideas

Python, FastAPI, Uvicorn, Google Gemini / Google GenAI Python SDK, Jinja2, HTML5, CSS3, and python-dotenv form the documented technology stack.

6. Expected Outcome

The idea establishes a foundation for a practical educational prototype and can later be extended with multilingual learning, voice interaction, mobile applications, LMS integration, progress tracking, and adaptive recommendations.

Prepared for project documentation and academic demonstration.